

22) If the equation $Hx=c$ is inconsistent for some c in $\mathbb{R}^n$, what can you say about the equation $Hx=0$? Why?

Solution: By the defn. of inconsistency, the equation $Hx=c$ has no solution for some c in $\mathbb{R}^n$. This is the negation of the statement that the equation $Hx=c$ has at least one solution for each c in $\mathbb{R}^n$. From the equivalence of the statements (g) and (d) in the Invertible Matrix Theorem, we can conclude that the equation $Hx=0$ has a non-trivial solution. $\square$

23) If an $n \times n$ matrix K cannot be row reduced to I_n , what can you say about the columns of K ? Why?

Solution: From the Invertible Matrix Theorem and the equivalence of the statements (b), (e) and (h), we conclude that the columns of K form a linearly dependent set and they do not span $\mathbb{R}^n$. $\square$

24) If L is $n \times n$ and the equation $Lx=0$ has the trivial solution, do the columns of L span $\mathbb{R}^n$? Why?

Solution: By The Invertible Matrix Theorem and the equivalence of the statements (d) and (h) we know that $Lx=0$ has only the trivial soln. if and only if the columns of L span $\mathbb{R}^n$. $Lx=0$ always has the trivial solution. But we don't know whether it has the nontrivial solution from the given conditions. $\therefore$ We cannot conclude whether the columns of L span $\mathbb{R}^n$ or not. $\square$

25) Verify the boxed statement preceding Example 1.

Solution: ~~Let A and B be square matrices.~~

Claim: Let A and B be square matrices. If $AB=I$, then A and B are both invertible, with $B=A^{-1}$ and $A=B^{-1}$.

Proof: By the Invertible Matrix Theorem and the equivalence of the statements (j) and (a), B is invertible. By the defn. of inverse, there exists a $n \times n$ matrix B^{-1} s.t. $BB^{-1}=B^{-1}B=I_n$.
 $AB=I \Rightarrow (AB)B^{-1}=IB^{-1} \Rightarrow A(BB^{-1})=IB^{-1}$ (Theorem 2(a))
 $\Rightarrow AI_n=I_nB^{-1}$ (by the defn. of inverse)
 $\Rightarrow A=B^{-1}$ (Theorem 2(e))